PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	Make **2-3 notes** of specific business requirements that will be analyzed. • *Will the app process transactions?* • *Does it do a lot of back-end processing?* • *Are there industry regulations that need to be considered?* *The app will process transactions by having several payment options. The app does require back-end processing including the ability of buyers to message the sellers and sales should be clear and quick to process. Payment handling regulations need to be considered to avoid any legal issues.*
II. Define the technical scope	List of technologies used by the application: • *Application programming interface (API)* • *Public key infrastructure (PKI)* • *SHA-256* • *SQL* Write **2-3 sentences** (40-60 words) that describe why you choose to prioritize that technology over the others. I consider SHA-256 to have the highest priority since it will be used to protect sensitive user data like passwords and credit card numbers. It involves the use of hash functions which are considered to have the highest encryption since once hashed, it will be difficult to decrypt it.
III. Decompose application	Sample data flow diagram
IV. Threat analysis	List **2 types of threats** in the PASTA worksheet that are risks to the information being handled by the application. • *What are the internal threats?* • *What are the external threats?* *Internal threats can include an employee with authorized access to the system could misuse their privileges and gain sensitive user*

	Information like credit card numbers. *External threats can include a threat actor could attempt a phishing campaign to gain sensitive information from app users.*
V. Vulnerability analysis	List **2 vulnerabilities** in the PASTA worksheet that could be exploited. • *Could there be things wrong with the codebase?* • *Could there be weaknesses in the database?* • *Could there be flaws in the network?* *An attacker could attempt to exploit the app database by SQL injections to gain sensitive information.* *Another vulnerability an attacker could exploit is that if the app has weak password requirements, then a brute force attack could be used to guess user login credentials.*
VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	List **4 security controls** that you've learned about that can reduce risk. • Applying principle of least privilege • Use of strong passwords • Having MFA enabled • Having prepared statements to mitigate SQL injections